CHAPTER 9

Excitons, Biexcitons and Trions

Contents

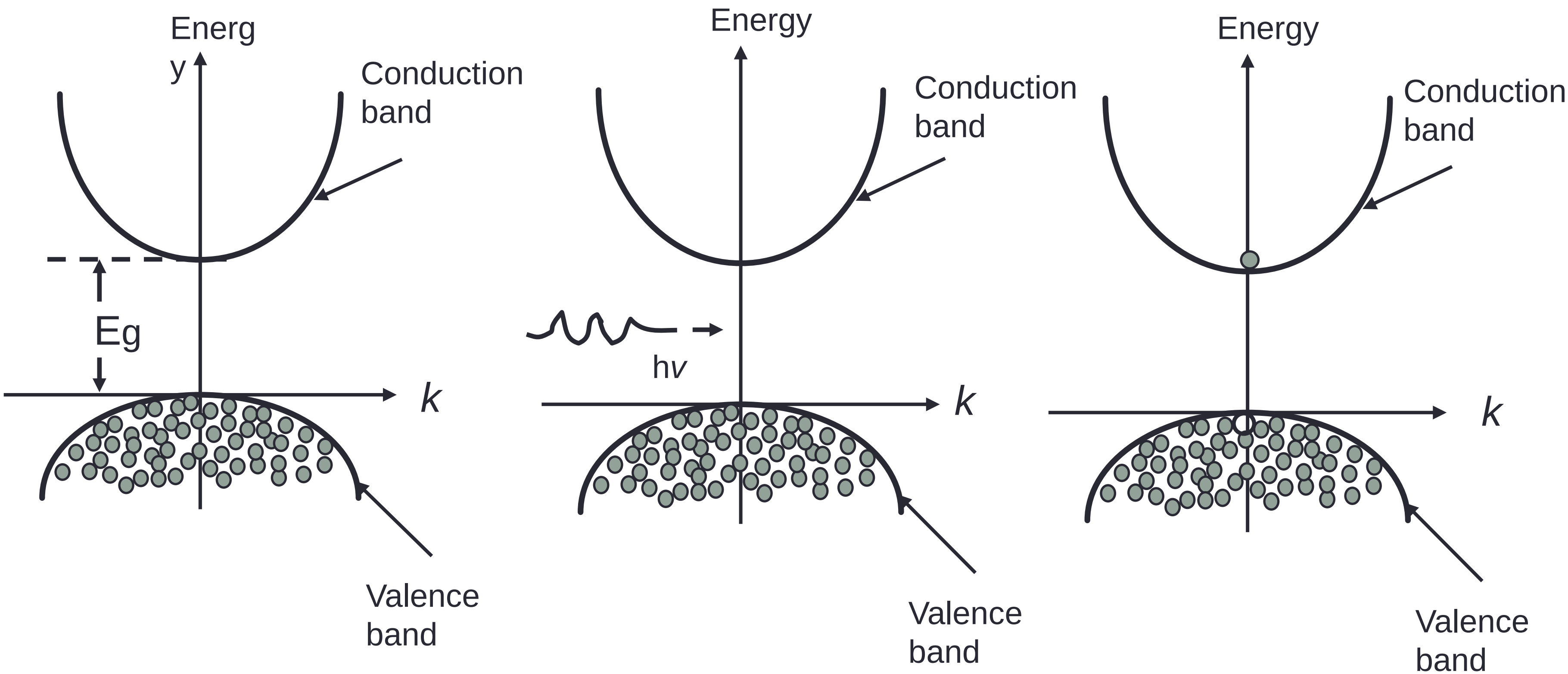
- In this chapter we introduce the concepts of Exciton, Biexciton and Trion.
- We discuss their behaviors in bulk semiconductors and semiconductors with reduced dimensionality.

Exciton

Ground state: Valence band is completely full, conduction is empty

Excited state: Excite electrons from valence band to conduction band by absorption of photons

Exciton: A bound state of an electron and a hole formed in an insulator or semiconductor through Coulomb interaction



9.1 Wannier and Frenkel Excitons

- Fig.9.1a: Coulomb interaction: with a Coulomb potential term $-e^2/\left(4\pi\varepsilon_0\varepsilon\left|\mathbf{r}_e-\mathbf{r}_h\right|\right)$
- Fig.9.1b: The dispersion relation of excitons :

$$E_{ex}(n_B,\mathbf{K}) = E_g - Ry^*\frac{1}{n_B^2} + \frac{\hbar^2\mathbf{K}^2}{2M}$$

with

$$n_B = 1,2,3\ldots \quad principal \quad quantum \quad number$$

$$Ry^* = 13.6eV\,\frac{\mu}{m_0}\frac{1}{\varepsilon^2} \quad exciton \quad Rydberg \quad energy$$

$$M = m_e + m_h,\ \mathbf{K} = \mathbf{k}_e + \mathbf{k}_h \quad translational \quad mass \quad and \quad wave \quad vector \quad of \quad the \quad exciton$$

$$\mu = \frac{m_e m_h}{m_e + m_h} \quad reduced \quad exciton \quad mass$$

$$a_B^{ex} = a_B^H \varepsilon \frac{m_0}{\mu} \quad \text{excitonic Bohr radius}.$$

Using the material parameters for typical semiconductors :

$$1meV \le Ry^* \le 200meV \ll E_g$$

$$50nm \ge \alpha_B \ge 1nm > \alpha_{lattice}$$

Excitons
- Wannier Excitons
- Frenkel Excitons

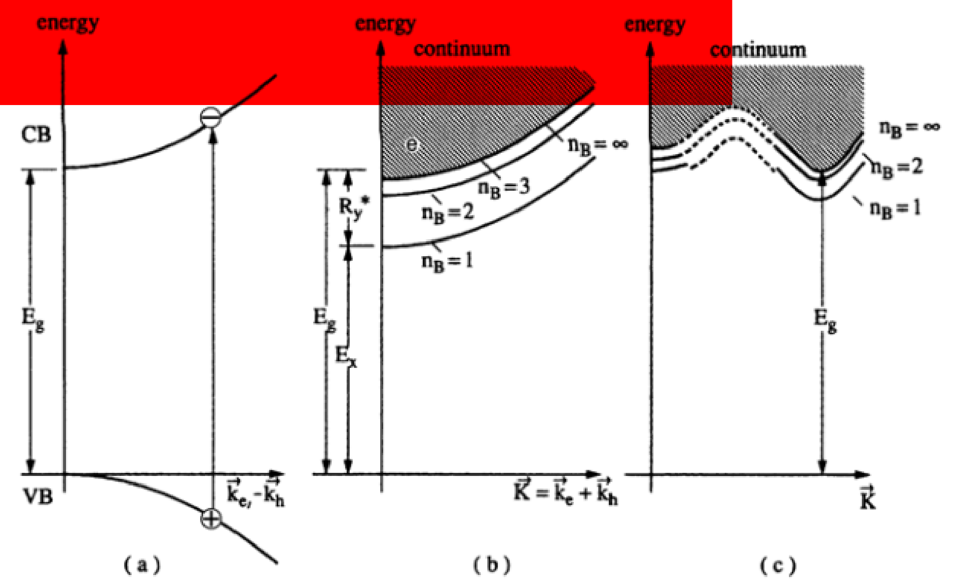


Fig. 9.1. A pair excitation in the scheme of valence and conduction band (**a**) in the exciton picture for a direct (**b**) and for an indirect gap semiconductor (**c**)

- For Wannier excitons , the exciton wavefunction:

$$\phi\left(\boldsymbol{K}, n_B, l, m\right) = \Omega^{-1/2} e^{i\boldsymbol{K}\cdot\boldsymbol{R}} \phi_e\left(\boldsymbol{r}_e\right) \phi_h\left(\boldsymbol{r}_h\right) \phi^{env}_{n_B,l,m}\left(\boldsymbol{r}_e - \boldsymbol{r}_h\right) \quad (9.4a)$$

with the center of mass **R :** $\boldsymbol{R} = \left(m_e\boldsymbol{r}_e + m_h\boldsymbol{r}_h\right) / \left(m_e + m_h\right)$

For direct semiconductors : $f_{nB} \propto \left|H^D_{cv}\right|^2 \frac{1}{n_B^3}$ (9.5)

- The electron–hole pair, likewise the exciton, is said to be in a spin singlet state.
- Spin triplet state: S=1 (if the spin flips in the transition), situated energetically below the singlet state.
- In the picture of second quantization, $\alpha^+_{\boldsymbol{k}_e}$、$\beta^+_{\boldsymbol{k}_h}$ for the creation operators of electrons in the conduction band and holes in the valence band.

  so creation operators for electron hole pairs : $\alpha^+_{\boldsymbol{k}_e}\beta^+_{\boldsymbol{k}_h}$

  The exciton creation operator : $B^+_{\boldsymbol{K}} = \sum_{\boldsymbol{k}'_e\boldsymbol{k}_h} \delta\left[\boldsymbol{K}\text{-}\left(\boldsymbol{k}_e+\boldsymbol{k}_h\right)\right] \alpha_{\boldsymbol{k}_e\boldsymbol{k}_h} \alpha^+_{\boldsymbol{k}_e}\beta^+_{\boldsymbol{k}_h}$ (9.6)

- Obey Bose commutation relations.
- In thermodynamic equilibrium for low densities: the excitons can be well described by Boltzmann statistics .
- For higher densities: they deviate more and more from ideal bosons until they end up in an electron–hole plasma made up entirely from fermions.

9.2 Corrections to the Simple Exciton Model

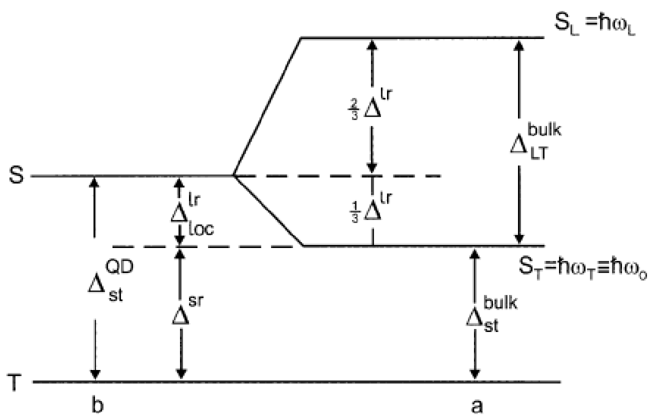
- A first group of corrections relevant for ε and μ .
- For GaAs : $E_{ex}^{b} < \mathrm{h}\omega_{LO};\quad a_B > a_{Pol} \Rightarrow \varepsilon = \varepsilon_s$
- For CdS, ZnO……: $E_{ex}^{b} \geq \mathrm{h}\omega_{LO};\quad a_B\ ;\ a_{Pol} \rightarrow \varepsilon_s \geq \varepsilon \geq \varepsilon_b$

  So using Haken potential :

$$\frac{1}{\varepsilon\left(\boldsymbol{r}_{e,h}\right)} = \frac{1}{\varepsilon_b} - \left(\frac{1}{\varepsilon_b} - \frac{1}{\varepsilon_s}\right)\left(1 - \frac{\exp\left(-\boldsymbol{r}_{eh}/\boldsymbol{r}_e^P\right) + \exp\left(-\boldsymbol{r}_{eh}/\boldsymbol{r}_h^P\right)}{2}\right) \qquad (9.8)$$

◆An increase of the exciton binding energy with increasing E_g

Fig. 9.2 The splitting of the 1S exciton with the notation of the various contributions to exchange splitting for bulk samples (**a**) and in a quantum dot (**b**). The abreviations have the following meanings: S = singlet, T = triplet, Δ^{lr} = long range or nanoanalytic (for $\boldsymbol{k} \rightarrow 0$) part of the exchange interaction, Δ^{sr} = short range or analytic part of the exchange interaction [03G1]



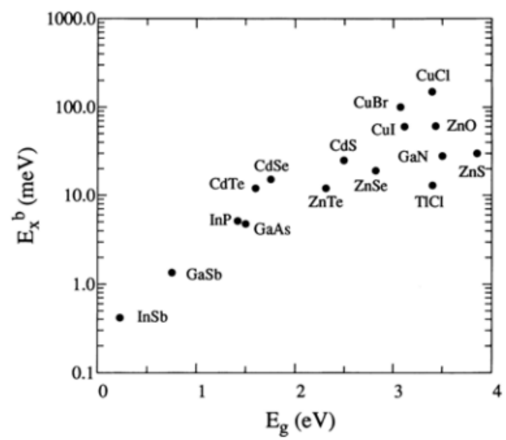


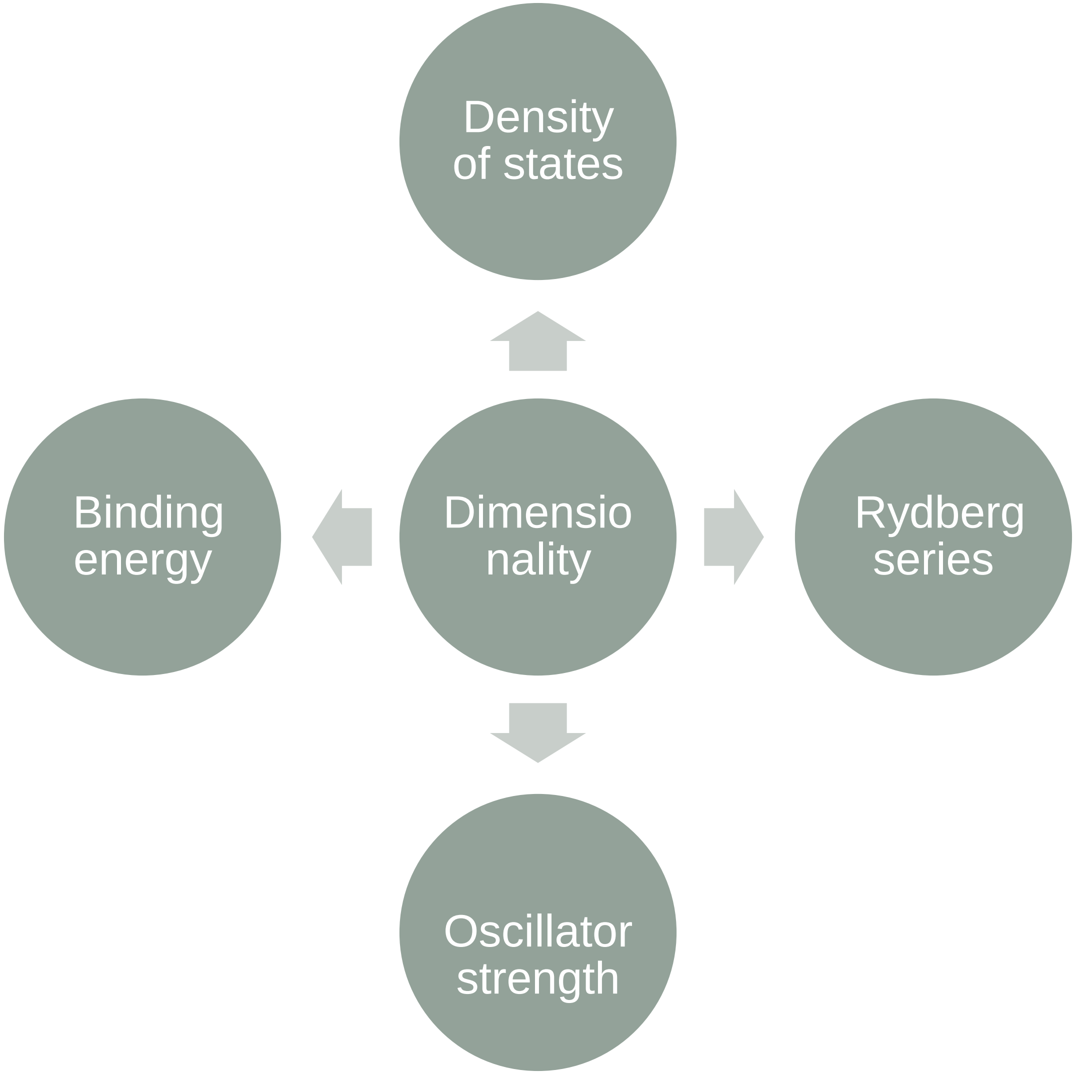
Fig. 9.3. The exciton binding energy E_{ex}^{b} as a function of the band-gap for various direct gap semiconductors ([82L1, 93H1, 93P1] of Chap. 1)

- Exchange interaction between electron and hole caused by their Coulomb interaction result in the splitting between singlet and triplet excitons Δ_{st} and the splitting of singlet state into a transverse and a longitudinal one Δ_{LT}
- The following relation holds for Wannier excitons:

$$\Delta_{st} = \Delta_{LT} \quad with \quad 0.1meV \le \Delta_{LT} \le 15meV \qquad (9.9)$$

- The singlet-triplet splitting is even enhanced by the compression of the excitonic wave function in quantum wells and especially in quantum dots.
- The excitation of an optically allowed exciton is accompanied by a polarization.
- The excitons can also be formed with holes in deeper valence bands.

9.3 The Influence of Dimensionality



- We consider an exciton:

3D :

$$3d: E(\mathbf{K}, n_B) = E_g - Ry^* \frac{1}{n_B^2} + \frac{\hbar^2 (K_x^2 + K_y^2 + K_z^2)}{2M} \qquad (9.10a)$$

For the oscillator strength f :

$$f(n_B) \propto n_B^{-3}; \quad a_B \propto a_B^H n_B; \quad n_B = 1, 2, 3K \qquad (9.10b)$$

2D :

$$2d: E(\mathbf{K}, n_B) = E_g + E_Q - Ry^* \frac{1}{\left(n_B - \frac{1}{2}\right)^2} + \frac{\hbar^2 (K_x^2 + K_y^2)}{2M} \qquad (9.10c)$$

with E_Q quantization energy:

$$f(n_B) \propto \left(n_B - \frac{1}{2}\right)^{-3}; \quad a_B \propto a_B^H \left(n_B - \frac{1}{2}\right); \quad n_B = 1, 2, 3K \qquad (9.10d)$$

	3D →2D
n_B	$n_B \rightarrow (n_B - 1/2)$
Ry*	The same
Oscillator strength	Increases
Excitonic Bohr radius	Decreases

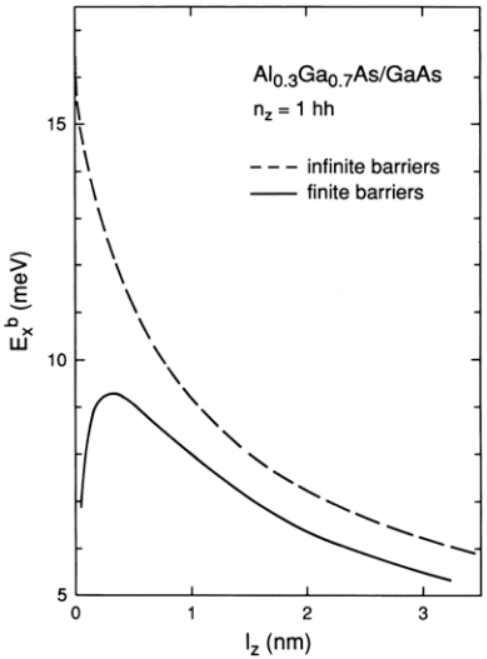


Fig. 9.4. The calculated binding energy of $n_z = 1$hh excitons in AlGaAs/GaAs quantum wells as a function of the well thickness l_z ([81M1, 84G1, 88K2])

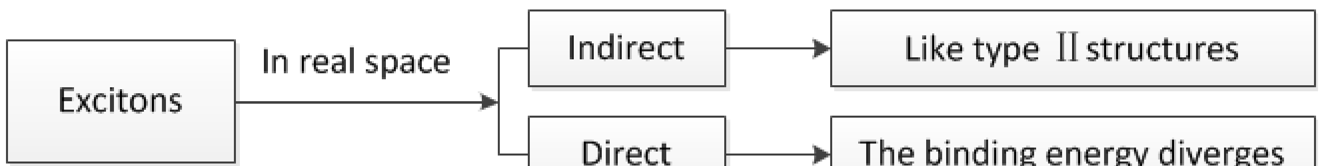
- The binding energy of the exciton in the quasi two-dimensional case：

$$E_{ex}^{b}(1S)=\frac{Ry^{*}}{\left(1+\frac{d_{eff}-3}{2}\right)^{2}} \quad with \quad d_{eff}=3-\exp\left(\frac{-Lw}{2a_{\beta}}\right) \tag{9.10}$$

The Sommerfeld factor F: describes the enhancement of the oscillator strength of the continuum states .

$$3d: F_{3d}=\frac{\pi}{W^{1/2}}\frac{e^{\pi W^{1/2}}}{\sinh\left(\pi W^{1/2}\right)} \quad with \quad W=\left(E-E_{g}\right)Ry^{*-1} \tag{9.11}$$

$$2d: F_{2d}=\frac{e^{\pi W^{1/2}}}{\cosh\left(\pi W^{1/2}\right)} \quad with \quad W=\left(E-E_{g}+E_{Q}\right)Ry^{*-1}$$

- 

- For quantum dots:

□ Weak confinement: $R\geq a_{B}, E_{Q}<Ry^{*}$

□ Medium or intermediate confinement : $a_{B}^{e}\geq R\geq a_{B}^{h}; E_{Q}\approx Ry^{*}, with \quad a_{B}^{e,h}=a_{B}^{H}\frac{m_{0}}{m_{e,h}}$

□ Strong confinement: $R\leq a_{B}^{h}, E_{Q}>Ry^{*}$

9.4 Biexitonns and Trions

- Biexiton or excitonic molecule： a new quasiparticle formed by two excitons.

  The dispersion relation is given :

$$E_{biex}(\mathbf{k}) = 2\left(E_g - E^b_{ex}\right) - E^b_{biex} + \frac{\hbar^2\mathbf{k}^2}{4M_{\mathrm{ex}}} \qquad (9.14)$$

Biexcitons also form bound states in quantum wells, wires and dots again in agreement with experiments.

- Trions: charged excitons or biexcitons, i.e., quasiparticles consisting of two electrons and one hole or vice versa.

  For quantum structures: trions also form bound states contributing to the luminescence below the free exciton energy.

9.5 Bound Exciton Complexes

Excitons can also be bound to defects.

- Shallow impurities.
  - The binding energy of an exciton is highest for a neutral acceptor, lower for a neutral donor and lower still for an ionized donor .
  - An ionized acceptor does not usually bind an exciton .
  - The absorption and emission lines of A^0X, D^0X and D^+X are often labelled I_1 , I_2 and I_3 lines, respectively.
  - The binding energy of an exciton to a neutral donor (acceptor) is usually much smaller than the binding energy of an electron (hole) to the donor (acceptor).
- Donor–acceptor pairs can be considered as “polycentric” bound excitons.
- One center can, under certain conditions, bind several excitons.
  - In indirect semiconductors
  - In quantum wells

9.6 Excitons in Disordered Systems

We consider the two-particle complex exciton localized in a disordered semiconductor.

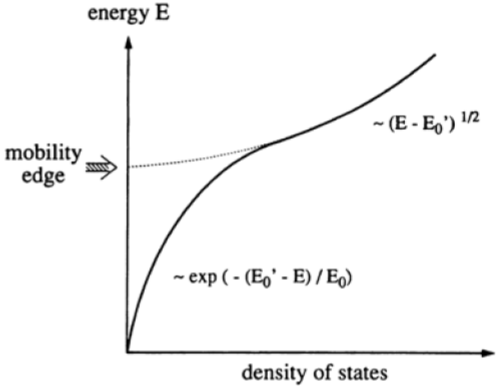


Fig. 9.5. The density of exciton states for $n_B = 1$ in a disordered semiconductor

- Low energies: start with excitons for which one carrier is localized and the other bound to it by Coulomb attraction.
- With increasing energy: a continuous transition to excitons localized as a whole . The mobility "edge".

- Features:
  - Localized excitons more frequently in those ionic semiconductors with alloying in the anions that form the valence band like $CdS_{1-x}Se_x$ or $ZnSe_{1-x}Te_x$ than in cation-substituted materials such as $Zn_{1-y}Cd_ySe$.
  - The exciton averages over a larger volume, thus diminishing the effective fluctuation and reducing the tail of localized exciton states.
- In quantum wells various types of disorder can contribute to the formation of tails of localized exciton states as already discussed for one-particle states:
  - Alloy disorder if the well and/or the barrier material is an alloy.
  - Interface roughness, i.e., well-width fluctuations.